

Sathiya Sudan

AI Engineer · GenAI, Computer Vision & ML

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SUMMARY

AI Engineer with 5+ years across machine learning, data science and generative AI. Currently building production AI for a leading Indian private bank — multi-agent assistants over MCP, signature and face verification, liveness detection, V-KYC audit automation and voice AI. MSc Data Science (Distinction), University of Essex.

EXPERIENCE

AI Engineer · [Company Name] — Client: leading Indian private bank Aug 2025 – Present

- **AI Banking:** multi-agent assistant — host planner plus 6 specialist agents (balance, transfers, FDs, statements, spend) — calling core banking APIs through an MCP server, secured with JWT/OAuth.
- **VerifyX:** automated V-KYC video audits with Whisper (ONNX Runtime) + GPT-4o compliance checks — 100% session coverage, 90% less audit time, flags customer-coaching fraud.
- **Syahi:** signature verification pipeline (YOLOX detection → Pix2Pix cleaning → MobileNetV2 matching) at 97% precision, ~1s per check.
- **Face verification & liveness:** selfie-vs-Aadhaar matching with VGGFace2 embeddings and MediaPipe — 98.4% accuracy, 95.2% precision; DeepPixBiS anti-spoofing at 0% HTER (Replay-Mobile), 0.42% ACER (OULU-NPU).
- **VEDA:** speech-to-speech voice agent (Whisper v3 → Llama-3.1-8B → HiFi-GAN) removing STT/TTS hops for lower latency.

Senior Software Engineer (GenAI) · Zensar Technologies — Client: Cisco Webex Apr 2025 – Jul 2025

- Built a LangGraph agent for enterprise sales insights (bookings, pipeline forecasts) with intent routing, validated tool calls, memory checkpointing on Azure Databricks and LangSmith tracing to control hallucinations.

Freelance AI / ML Engineer Mar 2023 – Feb 2025

- Delivered end-to-end LLM chatbots, RAG pipelines and recommendation systems.

Data Science Intern · University of Essex, UK Jan 2022 – Oct 2022

- XGBoost offer-eligibility model (tuned via GridSearchCV) deployed with FastAPI in a decision-support tool; contributed to an 82% rise in course-completion rate.

Software Engineer — Data Science / ML · BSH Home Appliances Feb 2018 – Sep 2021

- Built a collaborative-filtering recommendation engine and a real-time fraud detector (Isolation Forest, One-Class SVM) on Azure Databricks.
- Delivered RFM customer segmentation with a Streamlit insights app, deployed on Azure ML with PySpark.

SKILLS

GenAI & Agents	LangGraph, LangChain, multi-agent systems, MCP, RAG, GPT-4o, Llama 3.1, prompt engineering, LangSmith, vector DBs
Computer Vision	YOLOX, Pix2Pix, MobileNetV2, VGGFace2, MediaPipe, DeepPixBiS, OpenCV, face verification, anti-spoofing
Speech & NLP	Whisper, speech-to-speech, HiFi-GAN, ONNX Runtime, BERT, LSTM
ML & Data Science	PyTorch, scikit-learn, XGBoost, Random Forest, SVM, clustering, anomaly detection, A/B testing, Bayesian statistics
Engineering & Cloud	Python, SQL, R, FastAPI, Streamlit, Docker, Azure ML, Azure AI Foundry, Azure Databricks, PySpark, Git

EDUCATION

MSc Data Science — Distinction, University of Essex, UK · Academic Excellence International Scholarship 2021 – 2023

B.Tech, Information Technology, Panimalar Institute of Technology 2013 – 2017